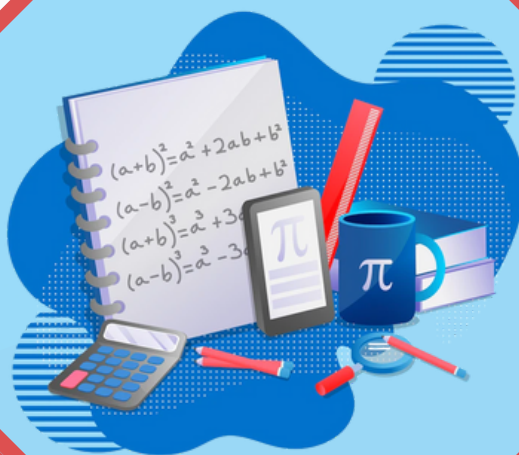


JEE (MAIN+ADVANCED)

Mathematics

ENTHUSIAST COURSE



STUDY MATERIAL

Limit, Continuity & Differentiability

English Medium

Copyright Statement

All rights including trademark and copyrights and rights of translation etc. reserved and vested exclusively with ALLEN Career Institute Private Limited. (ALLEN)

No part of this work may be copied, reproduced, adapted, abridged or translated, transcribed, transmitted, stored or distributed in any form retrieval system, computer system, photographic or other system or transmitted in any form or by any means whether electronic, magnetic, chemical or manual, mechanical, digital, optical, photocopying, recording or otherwise, or stored in any retrieval system of any nature without the written permission of the Allen Career Institute Private Limited. Any breach will entail legal action and prosecution without further notice.

This work is sold/distributed by Allen Career Institute Private Limited subject to the condition and undertaking given by the student that all proprietary rights (under the Trademark Act, 1999 and Copyright Act, 1957) of the work shall be exclusively belong to ALLEN Career Institute Private Limited. Neither the Study Materials and/or Test Series and/or the contents nor any part thereof i.e. work shall be reproduced, modify, re-publish, sub-license, upload on website, broadcast, post, transmit, disseminate, distribute, sell in market, stored in a retrieval system or transmitted in any form or by any means for reproducing or making multiple copies of it.

Any person who does any unauthorised act in relation to this work may be liable to criminal prosecution and civil claims for damages. Any violation or infringement of the propriety rights of Allen shall be punishable under Section- 29 & 52 of the Trademark Act, 1999 and under Section- 51, 58 & 63 of the Copyright Act, 1957 and any other Act applicable in India. All disputes are subjected to the exclusive jurisdiction of courts, tribunals and forums at Kota, Rajasthan only.

Note:- This publication is meant for educational and learning purposes. All reasonable care and diligence have been taken while editing and printing this publication. ALLEN Career Institute Private Limited shall not hold any responsibility for any error that may have inadvertently crept in. ALLEN Career Institute Private Limited is not responsible for the consequences of any action taken on the basis of this publication.

LIMIT

1. INTRODUCTION :

The concept of limit of a function is one of the fundamental ideas that distinguishes calculus from algebra and trigonometry. We use limits to describe the way a function f varies. Some functions vary continuously; small changes in x produce only small changes in $f(x)$. Other functions can have values that jump or vary erratically. We also use limits to define tangent lines to graphs of functions. This geometric application leads at once to the important concept of derivative of a function.

2. DEFINITION :

Let $f(x)$ be defined on an open interval about 'a' except possibly at 'a' itself. If $f(x)$ gets arbitrarily close to L (a finite number) for all x sufficiently close to 'a' we say that $f(x)$ approaches the limit L as x approaches 'a' and we write $\lim_{x \rightarrow a} f(x) = L$ and say "the limit of $f(x)$, as x approaches a , equals L ".

This implies if we can make the value of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a .

3. LEFT HAND LIMIT AND RIGHT HAND LIMIT OF A FUNCTION :

The value to which $f(x)$ approaches, as x tends to 'a' from the left hand side ($x \rightarrow a^-$) is called left hand limit of $f(x)$ at $x = a$. Symbolically, $LHL = \lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h)$.

The value to which $f(x)$ approaches, as x tends to 'a' from the right hand side ($x \rightarrow a^+$) is called right hand limit of $f(x)$ at $x = a$. Symbolically, $RHL = \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$.

Limit of a function $f(x)$ is said to exist as, $x \rightarrow a$ when $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = \text{Finite quantity}$.

Example :

Graph of $y = f(x)$

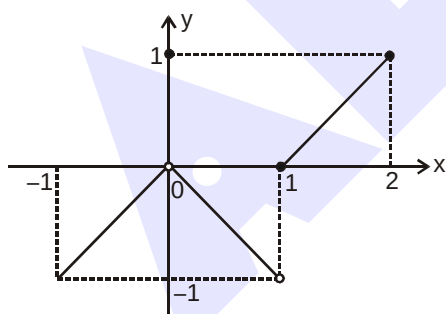


Fig. 1

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{h \rightarrow 0} f(-1+h) = f(-1^+) = -1$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(0-h) = f(0^-) = 0$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(0+h) = f(0^+) = 0$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{h \rightarrow 0} f(1-h) = f(1^-) = -1$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{h \rightarrow 0} f(1+h) = f(1^+) = 0$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{h \rightarrow 0} f(2-h) = f(2^-) = 1$$

$$\lim_{x \rightarrow 0} f(x) = 0 \text{ and } \lim_{x \rightarrow 1} f(x) \text{ does not exist.}$$

Important note :

In $\lim_{x \rightarrow a} f(x)$, $x \rightarrow a$ necessarily implies $x \neq a$. That is while evaluating limit at $x = a$, we are not concerned with the value of the function at $x = a$. In fact the function may or may not be defined at $x = a$.

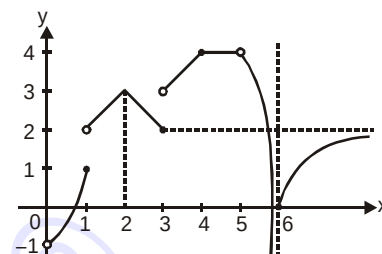
Also it is necessary to note that if $f(x)$ is defined only on one side of ' $x = a$ ', one sided limits are good enough to establish the existence of limits, & if $f(x)$ is defined on either side of ' a ' both sided limits are to be considered.

As in $\lim_{x \rightarrow 1} \cos^{-1} x = 0$, though $f(x)$ is not defined for $x > 1$, even in it's immediate vicinity.

Illustration 1 : Consider the adjacent graph of $y = f(x)$

Find the following :

- (a) $\lim_{x \rightarrow 0^-} f(x)$ (b) $\lim_{x \rightarrow 0^+} f(x)$ (c) $\lim_{x \rightarrow 1^-} f(x)$
 (d) $\lim_{x \rightarrow 1^+} f(x)$ (e) $\lim_{x \rightarrow 2^-} f(x)$ (f) $\lim_{x \rightarrow 2^+} f(x)$
 (g) $\lim_{x \rightarrow 3^-} f(x)$ (h) $\lim_{x \rightarrow 3^+} f(x)$ (i) $\lim_{x \rightarrow 4^-} f(x)$
 (j) $\lim_{x \rightarrow 4^+} f(x)$ (k) $\lim_{x \rightarrow \infty} f(x) = 2$ (l) $\lim_{x \rightarrow 6^-} f(x) = -\infty$



Solution : (a) As $x \rightarrow 0^-$: limit does not exist (the function is not defined to the left of $x = 0$)

(b) As $x \rightarrow 0^+ : f(x) \rightarrow -1 \Rightarrow \lim_{x \rightarrow 0^+} f(x) = -1$. (c) As $x \rightarrow 1^- : f(x) \rightarrow 1 \Rightarrow \lim_{x \rightarrow 1^-} f(x) = 1$.

(d) As $x \rightarrow 1^+ : f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow 1^+} f(x) = 2$. (e) As $x \rightarrow 2^- : f(x) \rightarrow 3 \Rightarrow \lim_{x \rightarrow 2^-} f(x) = 3$.

(f) As $x \rightarrow 2^+ : f(x) \rightarrow 3 \Rightarrow \lim_{x \rightarrow 2^+} f(x) = 3$. (g) As $x \rightarrow 3^- : f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow 3^-} f(x) = 2$.

(h) As $x \rightarrow 3^+ : f(x) \rightarrow 3 \Rightarrow \lim_{x \rightarrow 3^+} f(x) = 3$. (i) As $x \rightarrow 4^- : f(x) \rightarrow 4 \Rightarrow \lim_{x \rightarrow 4^-} f(x) = 4$.

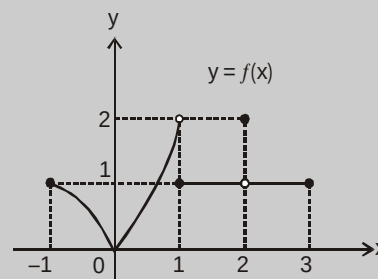
(j) As $x \rightarrow 4^+ : f(x) \rightarrow 4 \Rightarrow \lim_{x \rightarrow 4^+} f(x) = 4$. (k) As $x \rightarrow \infty : f(x) \rightarrow 2 \Rightarrow \lim_{x \rightarrow \infty} f(x) = 2$.

(l) As $x \rightarrow 6^- , f(x) \rightarrow -\infty \Rightarrow \lim_{x \rightarrow 6^-} f(x) = -\infty$ limit does not exist because it is not finite.

Do yourself - 1 :

1. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false ?

- (a) $\lim_{x \rightarrow -1^+} f(x) = 1$ (b) $\lim_{x \rightarrow 2} f(x)$ does not exist
 (c) $\lim_{x \rightarrow 2} f(x) = 2$ (d) $\lim_{x \rightarrow 1^-} f(x) = 2$
 (e) $\lim_{x \rightarrow 1} f(x)$ does not exist (f) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$
 (g) $\lim_{x \rightarrow c} f(x)$ exists at every $c \in (-1, 1)$
 (h) $\lim_{x \rightarrow c} f(x)$ exists at every $c \in (1, 3)$
 (i) $\lim_{x \rightarrow 1^-} f(x) = 0$ (j) $\lim_{x \rightarrow 3^+} f(x)$ does not exist.



4. FUNDAMENTAL THEOREMS ON LIMITS :

Let $\lim_{x \rightarrow a} f(x) = l$ & $\lim_{x \rightarrow a} g(x) = m$. If l & m exist finitely then :

- (a) Sum rule : $\lim_{x \rightarrow a} \{f(x) + g(x)\} = l + m$ (b) Difference rule : $\lim_{x \rightarrow a} \{f(x) - g(x)\} = l - m$
- (c) Product rule : $\lim_{x \rightarrow a} f(x) \cdot g(x) = l \cdot m$ (d) Quotient rule : $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{l}{m}$, provided $m \neq 0$
- (e) Constant multiple rule : $\lim_{x \rightarrow a} k f(x) = k \lim_{x \rightarrow a} f(x)$; where k is constant.
- (f) Power rule : If m and n are integers then $\lim_{x \rightarrow a} [f(x)]^{m/n} = l^{m/n}$ provided $l^{m/n}$ is a real number.
- (g) $\lim_{x \rightarrow a} f[g(x)] = f\left(\lim_{x \rightarrow a} g(x)\right) = f(m)$; provided $f(x)$ is continuous at $x = m$.

For example : $\lim_{x \rightarrow a} \ell n(g(x)) = \ell n[\lim_{x \rightarrow a} g(x)]$

$= \ell n(m)$; provided $\ell n x$ is continuous at $x = m$, $m = \lim_{x \rightarrow a} g(x)$.

5. INDETERMINATE FORMS :

$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 1^\infty, 0^0, \infty^0$$

Initially we will deal with first five forms only and the other two forms will come up after we have gone through differentiation.

Note : (i) Here $0, 1$ are not exact, infact both are approaching to their corresponding values.

(ii) We cannot plot ∞ on the paper. Infinity (∞) is a symbol & not a number It does not obey the laws of elementary algebra,

$$(a) \infty + \infty \rightarrow \infty \quad (b) \infty \times \infty \rightarrow \infty \quad (c) \infty^\infty \rightarrow \infty \quad (d) 0^\infty \rightarrow 0$$

6. GENERAL METHODS TO BE USED TO EVALUATE LIMITS :

(a) Factorization :

Important factors :

- (i) $x^n - a^n = (x - a)(x^{n-1} + ax^{n-2} + \dots + a^{n-1})$, $n \in \mathbb{N}$
- (ii) $x^n + a^n = (x + a)(x^{n-1} - ax^{n-2} + \dots + a^{n-1})$, n is an odd natural number.

Note : $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$

Illustration 2 : Evaluate : $\lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3 - 3x^2 + 2x} \right]$

Solution : We have

$$\begin{aligned} \lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x^3 - 3x^2 + 2x} \right] &= \lim_{x \rightarrow 2} \left[\frac{1}{x-2} - \frac{2(2x-3)}{x(x-1)(x-2)} \right] = \lim_{x \rightarrow 2} \left[\frac{x(x-1) - 2(2x-3)}{x(x-1)(x-2)} \right] \\ &= \lim_{x \rightarrow 2} \left[\frac{x^2 - 5x + 6}{x(x-1)(x-2)} \right] = \lim_{x \rightarrow 2} \left[\frac{(x-2)(x-3)}{x(x-1)(x-2)} \right] = \lim_{x \rightarrow 2} \left[\frac{x-3}{x(x-1)} \right] = -\frac{1}{2} \end{aligned}$$

Do yourself - 2 :

1. Evaluate: $\lim_{x \rightarrow 1} \frac{x-1}{2x^2-7x+5}$

2. What $\lim_{x \rightarrow 5} \frac{x^2-x-20}{x-5}$?

3. Find $\lim_{x \rightarrow 4} \frac{x^3-4x^2}{x^2-16}$

4. Evaluate $\lim_{x \rightarrow 1} \frac{x^3-1}{x-1}$

5. Evaluate $\lim_{x \rightarrow \frac{1}{2}} \frac{4x^2-1}{2x-1}$

6. Evaluate $\lim_{x \rightarrow 2} \frac{x^3-6x^2+11x-6}{x^2-6x+8}$

7. Evaluate: $\lim_{x \rightarrow 3} \frac{x^3-7x^2+15x-9}{x^4-5x^3+27x-27}$

(A) 2/9

(B) 1/9

(C) -2/9

(D) None of these

8. Evaluate: $\lim_{x \rightarrow 2} \frac{x^3-2x-4}{x^2-3x+2}$

9. Evaluate: $\lim_{x \rightarrow a} \frac{x^3-a^3}{x^2-ax}$

10. $\lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right)$ is equal to

(A) -1

(B) 0

(C) 1

(D) D.N.E.

11. $\lim_{x \rightarrow 1} \frac{\sqrt[n]{x}-1}{\sqrt[m]{x}-1}$ (m and n integers) is equal to

(A) 0

(B) 1

(C) $\frac{m}{n}$ (D) $\frac{n}{m}$

12. $\lim_{x \rightarrow 1} \frac{\left(\sum_{k=1}^{100} x^k \right) - 100}{x-1}$

(b) Rationalization or double rationalization :

Illustration 3 : Evaluate: $\lim_{x \rightarrow 1} \frac{4-\sqrt{15x+1}}{2-\sqrt{3x+1}}$

Solution :

$$\lim_{x \rightarrow 1} \frac{4-\sqrt{15x+1}}{2-\sqrt{3x+1}} = \lim_{x \rightarrow 1} \frac{(4-\sqrt{15x+1})(2+\sqrt{3x+1})(4+\sqrt{15x+1})}{(2-\sqrt{3x+1})(4+\sqrt{15x+1})(2+\sqrt{3x+1})}$$

$$= \lim_{x \rightarrow 1} \frac{(15-15x)}{(3-3x)} \times \frac{2+\sqrt{3x+1}}{4+\sqrt{15x+1}} = \frac{5}{2}$$

Illustration 4 : Evaluate : $\lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2 + 8} - \sqrt{10 - x^2}}{\sqrt{x^2 + 3} - \sqrt{5 - x^2}} \right)$

Solution : This is of the form $\frac{3-3}{2-2} = \frac{0}{0}$ if we put $x = 1$

To eliminate the $\frac{0}{0}$ factor, multiply by the conjugate of numerator and the conjugate of the denominator

$$\begin{aligned} \therefore \text{Limit} &= \lim_{x \rightarrow 1} \left(\sqrt{x^2 + 8} - \sqrt{10 - x^2} \right) \frac{(\sqrt{x^2 + 8} + \sqrt{10 - x^2})}{(\sqrt{x^2 + 8} + \sqrt{10 - x^2})} \times \frac{(\sqrt{x^2 + 3} + \sqrt{5 - x^2})}{(\sqrt{x^2 + 3} + \sqrt{5 - x^2})(\sqrt{x^2 + 3} - \sqrt{5 - x^2})} \\ &= \lim_{x \rightarrow 1} \frac{\sqrt{x^2 + 3} + \sqrt{5 - x^2}}{\sqrt{x^2 + 8} + \sqrt{10 - x^2}} \times \frac{(x^2 + 8) - (10 - x^2)}{(x^2 + 3) - (5 - x^2)} = \lim_{x \rightarrow 1} \left(\frac{\sqrt{x^2 + 3} + \sqrt{5 - x^2}}{\sqrt{x^2 + 8} + \sqrt{10 - x^2}} \right) \times 1 = \frac{2+2}{3+3} = \frac{2}{3} \end{aligned}$$

Do yourself - 3 :

1. Evaluate : $\lim_{x \rightarrow 0} \frac{\sqrt{p+x} - \sqrt{p-x}}{\sqrt{q+x} - \sqrt{q-x}}$

2. Evaluate : $\lim_{x \rightarrow a} \frac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}$, $a \neq 0$

3. If $G(x) = -\sqrt{25-x^2}$, then find the $\lim_{x \rightarrow 1} \left(\frac{G(x) - G(1)}{x-1} \right)$

4. $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{2x}$ is equal to

(A) 0

(B) 1

(C) $\frac{1}{2}$

(D) $\frac{1}{4}$

5. $\lim_{x \rightarrow 2} \frac{\sqrt{1+\sqrt{2+x}} - \sqrt{3}}{x-2}$ is equal to

(A) $\frac{1}{\sqrt{3}}$

(B) $\sqrt{3}$

(C) $\frac{1}{4\sqrt{3}}$

(D) $\frac{1}{8\sqrt{3}}$

(c) Limit when $x \rightarrow \infty$:

(i) Divide by greatest power of x in numerator and denominator.

(ii) Put $x = 1/y$ and apply $y \rightarrow 0$

Illustration 5 : Evaluate : $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{3x^2 + 2x - 5}$

Solution : $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{3x^2 + 2x - 5}, \quad \left(\frac{\infty}{\infty} \text{ form} \right)$

Put $x = \frac{1}{y}$

Limit = $\lim_{y \rightarrow 0} \frac{1 + y + y^2}{3 + 2y - 5y^2} = \frac{1}{3}$

Illustration 6 : If $\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 2$, then

- (A) $a = 1, b = 1$ (B) $a = 1, b = 2$ (C) $a = 1, b = -2$ (D) none of these

Solution : $\lim_{x \rightarrow \infty} \left(\frac{x^3 + 1}{x^2 + 1} - (ax + b) \right) = 2 \Rightarrow \lim_{x \rightarrow \infty} \frac{x^3(1-a) - bx^2 - ax + (1-b)}{x^2 + 1} = 2$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x(1-a) - b - \frac{a}{x} + \frac{(1-b)}{x^2}}{1 + \frac{1}{x^2}} = 2 \Rightarrow 1 - a = 0, -b = 2 \Rightarrow a = 1, b = -2 \quad \text{Ans. (C)}$$

Do yourself - 4 :

1. Evaluate : $\lim_{n \rightarrow \infty} \frac{\lfloor n+2 \rfloor + \lfloor n+1 \rfloor}{\lfloor n+2 \rfloor - \lfloor n+1 \rfloor}$

2. Evaluate : $\lim_{n \rightarrow \infty} (n - \sqrt{n^2 + n})$

3. Evaluate $\lim_{n \rightarrow \infty} \frac{3(n+1)}{(n+1)^3 \lfloor 3n \rfloor}, n \in \mathbb{N}$

4. Find $\lim_{n \rightarrow \infty} \frac{\sqrt{n^2 + 1} + \sqrt{n}}{\sqrt[4]{n^3 + 1} - \sqrt{n}}$

5. Find $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2} \right) \left(1 - \frac{1}{3^2} \right) \left(1 - \frac{1}{4^2} \right) \dots \left(1 - \frac{1}{n^2} \right)$

6. Evaluate $\lim_{n \rightarrow \infty} \frac{1 \cdot n + 2 \cdot (n-1) + 3 \cdot (n-2) \dots + n \cdot 1}{n^3}$

7. If the value of $\lim_{n \rightarrow \infty} n^3 \left(1 - \frac{3}{7} \right) \left(1 - \frac{3}{8} \right) \left(1 - \frac{3}{9} \right) \dots \left(1 - \frac{3}{n} \right)$ is k then find the sum of digits of k .

8. $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 1} - \sqrt{x^2 - 1} \right) =$

- (A) 0 (B) 1 (C) 2 (D) None

9. $\lim_{n \rightarrow \infty} \frac{(n+1)^4 - (n-1)^4}{(n+1)^4 + (n-1)^4}$ is equal to

- (A) -1 (B) 0 (C) 1 (D) D.N.E.

(d) Squeeze play theorem (Sandwich theorem) :

Statement : If $f(x) \leq g(x) \leq h(x)$; $\forall x$ in the neighbourhood at $x = a$ and

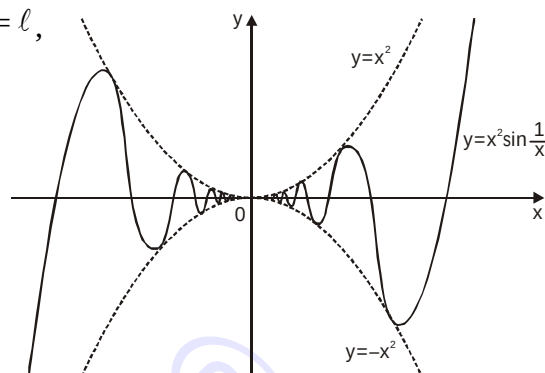
$$\lim_{x \rightarrow a} f(x) = \ell = \lim_{x \rightarrow a} h(x) \text{ then } \lim_{x \rightarrow a} g(x) = \ell,$$

Ex.1 $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0,$

$$\because \sin\left(\frac{1}{x}\right) \text{ lies between } -1 \text{ \& } 1$$

$$\Rightarrow -x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

$$\Rightarrow \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0 \text{ as } \lim_{x \rightarrow 0} (-x^2) = \lim_{x \rightarrow 0} x^2 = 0$$



Ex.2 $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$

$$\because \sin\left(\frac{1}{x}\right) \text{ lies between } -1 \text{ \& } 1$$

$$\Rightarrow -x \leq x \sin \frac{1}{x} \leq x$$

$$\Rightarrow \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0 \text{ as } \lim_{x \rightarrow 0} (-x) = \lim_{x \rightarrow 0} x = 0$$

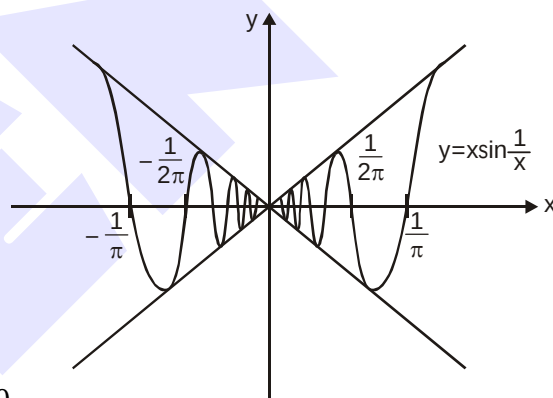


Illustration 7 : Evaluate : $\lim_{n \rightarrow \infty} \frac{[x] + [2x] + [3x] + \dots + [nx]}{n^2}$ (Where $[.]$ denotes the greatest integer function.)

Solution : We know that $x - 1 < [x] \leq x$

$$\Rightarrow x + 2x + \dots + nx - n < \sum_{r=1}^n [rx] \leq x + 2x + \dots + nx$$

$$\Rightarrow \frac{xn}{2}(n+1) - n < \sum_{r=1}^n [rx] \leq \frac{x \cdot n(n+1)}{2} \Rightarrow \frac{x}{2} \left(1 + \frac{1}{n}\right) - \frac{1}{n} < \frac{1}{n^2} \sum_{r=1}^n [rx] \leq \frac{x}{2} \left(1 + \frac{1}{n}\right)$$

$$\text{Now, } \lim_{n \rightarrow \infty} \frac{x}{2} \left(1 + \frac{1}{n}\right) = \frac{x}{2} \text{ and } \lim_{n \rightarrow \infty} \frac{x}{2} \left(1 + \frac{1}{n}\right) - \frac{1}{n} = \frac{x}{2}$$

$$\text{Thus, } \lim_{n \rightarrow \infty} \frac{[x] + [2x] + \dots + [nx]}{n^2} = \frac{x}{2}$$

7. LIMIT OF TRIGONOMETRIC FUNCTIONS :

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 = \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} \quad [\text{where } x \text{ is measured in radians}]$$

(a) If $\lim_{x \rightarrow a} f(x) = 0$, then $\lim_{x \rightarrow a} \frac{\sin f(x)}{f(x)} = 1$, e.g. $\lim_{x \rightarrow 1} \frac{\sin(\ln x)}{(\ln x)} = 1$

Illustration 8 : Evaluate : $\lim_{x \rightarrow 0} \frac{x^3 \cot x}{1 - \cos x}$

Solution : $\lim_{x \rightarrow 0} \frac{x^3 \cos x}{\sin x (1 - \cos x)} = \lim_{x \rightarrow 0} \frac{x^3 \cos x (1 + \cos x)}{\sin x \cdot \sin^2 x} = \lim_{x \rightarrow 0} \frac{x^3}{\sin^3 x} \cdot \cos x (1 + \cos x) = 2$

Illustration 9 : Evaluate : $\lim_{x \rightarrow 0} \frac{(2+x)\sin(2+x) - 2\sin 2}{x}$

Solution :
$$\lim_{x \rightarrow 0} \frac{2(\sin(2+x) - \sin 2) + x \sin(2+x)}{x} = \lim_{x \rightarrow 0} \left(\frac{2 \cdot 2 \cdot \cos\left(2 + \frac{x}{2}\right) \sin \frac{x}{2}}{x} + \sin(2+x) \right)$$

$$= \lim_{x \rightarrow 0} \frac{2 \cos\left(2 + \frac{x}{2}\right) \sin \frac{x}{2}}{\frac{x}{2}} + \lim_{x \rightarrow 0} \sin(2+x) = 2\cos 2 + \sin 2$$

Illustration 10 : Evaluate : $\lim_{n \rightarrow \infty} \frac{\sin \frac{a}{n}}{\tan \frac{b}{n+1}}$

Solution : As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$ and $\frac{a}{n}$ also tends to zero

$\sin \frac{a}{n}$ should be written as $\frac{\sin \frac{a}{n}}{\frac{a}{n}}$ so that it looks like $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$

The given limit
$$= \lim_{n \rightarrow \infty} \left(\frac{\sin \frac{a}{n}}{\frac{a}{n}} \right) \left(\frac{\frac{b}{n+1}}{\tan \frac{b}{n+1}} \right) \cdot \frac{a(n+1)}{n \cdot b}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{\sin \frac{a}{n}}{\frac{a}{n}} \right) \left(\frac{\frac{b}{n+1}}{\tan \frac{b}{n+1}} \right) \cdot \frac{a}{b} \left(1 + \frac{1}{n} \right) = 1 \times 1 \times \frac{a}{b} \times 1 = \frac{a}{b}$$

Do yourself - 5 :

1. Evaluate $\lim_{x \rightarrow 0} \frac{\sin \alpha x}{\tan \beta x}$

2. Evaluate $\lim_{x \rightarrow y} \frac{\sin^2 x - \sin^2 y}{x^2 - y^2}$

3. Evaluate $\lim_{h \rightarrow 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h}$

4. Evaluate $\lim_{x \rightarrow 0} \left[\frac{-2x}{\tan x} \right]$, where $[.]$ denotes greatest integer function.

5. $\lim_{x \rightarrow 1} \frac{\cos(\pi x / 2)}{1 - x} =$

- (A) 0 (B) π (C) $\pi/2$ (D) 2π

6. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x) \sin 5x}{x^2 \sin 3x} =$

- (A) $10/3$ (B) $3/10$ (C) $6/5$ (D) $5/6$

7. Evaluate $\lim_{x \rightarrow 1^-} \frac{1 - \sqrt{x}}{(\cos^{-1} x)^2}$

8. $\lim_{x \rightarrow 0} \frac{1 - \cos 5x}{3x^2}$

9. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tan x}{1 - \sqrt{2} \sin x}$

10. $\lim_{x \rightarrow 0} \frac{64}{x^8} \left[1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right]$

11. Using Sandwich theorem, evaluate

$$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2 + 1}} + \frac{1}{\sqrt{n^2 + 2}} + \dots + \frac{1}{\sqrt{n^2 + 2n}} \right)$$

8. LIMIT OF EXPONENTIAL FUNCTIONS :

(a) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ell \ln a (a > 0)$ In particular $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$.

In general if $\lim_{x \rightarrow a} f(x) = 0$, then $\lim_{x \rightarrow a} \frac{a^{f(x)} - 1}{f(x)} = \ell \ln a, a > 0$

Illustration 11 : Evaluate : $\lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x}$

Solution : $\lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x} = \lim_{x \rightarrow 0} \frac{e^x \times e^{(\tan x - x)} - e^x}{\tan x - x}$

$$= \lim_{x \rightarrow 0} \frac{e^x (e^{\tan x - x} - 1)}{\tan x - x} = \lim_{x \rightarrow 0} \frac{e^x (e^y - 1)}{y} \text{ where } y = \tan x - x \text{ and } \lim_{y \rightarrow 0} \frac{e^y - 1}{y} = 1$$

$$= e^0 \times 1 \quad [\text{as } x \rightarrow 0, \tan x - x \rightarrow 0]$$

$$= 1 \times 1 = 1$$

Do yourself - 6 :

1. Evaluate : $\lim_{x \rightarrow a} \frac{e^x - e^a}{x - a}$

2. Evaluate : $\lim_{x \rightarrow 0} \frac{2^x - 1}{(1+x)^{1/2} - 1}$

3. Evaluate $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x/2}$

4. Evaluate $\lim_{x \rightarrow a} \frac{a^x - a^a}{x - a}, a > 0.$

5. $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x}{x^2}$

6. $\lim_{x \rightarrow 0} \frac{e^{\sin 2x} - e^{\sin x}}{x}$

7. $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2}$

8. Find $\lim_{n \rightarrow \infty} \frac{n!}{(n+1)! - n!}$

(b) (i) $\lim_{x \rightarrow 0} (1+x)^{1/x} = e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$ (Note : The base and exponent depends on the same

variable.) **In general**, if $\lim_{x \rightarrow a} f(x) = 0$, then $\lim_{x \rightarrow a} (1+f(x))^{1/f(x)} = e$

(ii) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$

(iii) If $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} \phi(x) = \infty$, then ; $\lim_{x \rightarrow a} [f(x)]^{\phi(x)} = e^k$

where $k = \lim_{x \rightarrow a} \phi(x) [f(x) - 1]$

Illustration 12 : Evaluate $\lim_{x \rightarrow 1} (\log_3 3x)^{\log_x 3}$

Solution : $\lim_{x \rightarrow 1} (\log_3 3x)^{\log_x 3} = \lim_{x \rightarrow 1} (\log_3 3 + \log_3 x)^{\log_x 3}$

$$= \lim_{x \rightarrow 1} (1 + \log_3 x)^{1/\log_3 x} = e \quad \because \log_b a = \frac{1}{\log_a b}$$

Illustration 13 : Evaluate : $\lim_{x \rightarrow 0} \frac{x \ln(1+2 \tan x)}{1 - \cos x}$

Solution : $\lim_{x \rightarrow 0} \frac{x \ln(1+2 \tan x)}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{x \ln(1+2 \tan x)}{\frac{1 - \cos x}{x^2} \cdot x^2} \cdot \frac{2 \tan x}{2 \tan x} = 4$

Illustration 14 : Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1}{2x^2 + 3} \right)^{4x^2 + 2}$ **Solution :** Since it is in the form of 1^∞

$$\lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1}{2x^2 + 3} \right)^{4x^2 + 2} = e^{\lim_{x \rightarrow \infty} \left(\frac{2x^2 - 1 - 2x^2 - 3}{2x^2 + 3} \right) (4x^2 + 2)} = e^{-8}$$

1. Evaluate : $\lim_{x \rightarrow \infty} x\{\ell n(x+a) - \ell nx\}$
2. Evaluate : $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{pn+q}$
3. Evaluate : $\lim_{x \rightarrow 0} \left(1 + \tan^2 \sqrt{x}\right)^{\frac{1}{2x}}$
4. Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1}\right)^{x+4}$

(c) If $\lim_{x \rightarrow a} f(x) = A > 0$ & $\lim_{x \rightarrow a} \phi(x) = B$, then $\lim_{x \rightarrow a} [f(x)]^{\phi(x)} = e^{B \ln A} = A^B$

Illustration 15: Evaluate: $\lim_{x \rightarrow \infty} \left(\frac{7x^2 + 1}{5x^2 - 1} \right)^{\frac{x^5}{1-x^3}}$

Solution : Here $f(x) = \frac{7x^2 + 1}{5x^2 - 1}$, $\phi(x) = \frac{x^5}{1 - x^3} = \frac{x^2 \cdot x^3}{1 - x^3} = \frac{x^2}{\frac{1}{x^3} - 1}$

$$\therefore \lim_{x \rightarrow \infty} f(x) = \frac{7}{5} \quad \& \quad \lim_{x \rightarrow \infty} \phi(x) \rightarrow -\infty$$

$$\Rightarrow \lim_{x \rightarrow \infty} (f(x))^{\phi(x)} = \left(\frac{7}{5}\right)^{-\infty} = 0$$

1. Evaluate : $\lim_{x \rightarrow \infty} \left(\frac{1+5x^2}{1+3x^2} \right)^{-x^2}$
2. Evaluate $\lim_{n \rightarrow \infty} (4^n + 5^n)^{1/n}$
3. Evaluate $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x - 1}{3x^2 + 2x + 4} \right)^{\frac{3x^2 + x}{x - 2}}$
4. If $f(x) = \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x$ then find $\lim_{x \rightarrow \infty} f(x)$
5. Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\frac{\sin x}{x}}{1 - \frac{\sin x}{x}}}$
6. Evaluate $\lim_{x \rightarrow 0} (1 + x)^{\operatorname{cosec} x}$
7. Evaluate $\lim_{x \rightarrow 0} (\cos x)^{\cot x}$
8. If x_1 and x_2 are the real and distinct roots of $ax^2 + bx + c = 0$ then find $\lim_{x \rightarrow x_1} \left(1 + \sin(ax^2 + bx + c) \right)^{\frac{1}{x - x_1}}$
9. Evaluate $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{2/x}$; $(a, b, c > 0)$
10. Evaluate (a) $\lim_{x \rightarrow 0} (1 - 2x)^{1/x}$; (b) $\lim_{x \rightarrow 1} x^{\cot \pi x}$
11. If $\lim_{x \rightarrow \infty} \left[\frac{2x^2 + 3}{2x^2 + 5} \right]^{8x^2 + 3} = e^{-k}$, then the value of k is

9. LIMIT USING SERIES EXPANSION :

Expansion of function like binomial expansion, exponential & logarithmic expansion, expansion of $\sin x$, $\cos x$, $\tan x$ should be remembered by heart which are given below :

$$(a) \quad a^x = 1 + \frac{x \ln a}{1!} + \frac{x^2 \ln^2 a}{2!} + \frac{x^3 \ln^3 a}{3!} + \dots, x \in \mathbb{R}, a > 0, a \neq 1$$

$$(b) \quad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, x \in \mathbb{R}$$

$$(c) \quad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \text{ for } -1 < x \leq 1$$

$$(d) \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, x \in \mathbb{R}$$

$$(e) \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, x \in \mathbb{R}$$

$$(f) \quad \tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots, -\frac{\pi}{2} < x < \frac{\pi}{2}$$

$$(g) \quad \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots, x \in (-1, 1)$$

$$(h) \quad \sin^{-1} x = x + \frac{1^2}{3!} x^3 + \frac{1^2 \cdot 3^2}{5!} x^5 + \frac{1^2 \cdot 3^2 \cdot 5^2}{7!} x^7 + \dots, x \in (-1, 1)$$

$$(i) \quad \sec^{-1} x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + \dots, x \in (-\infty, -1) \cup (1, \infty)$$

$$(j) \quad (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots, n \in \mathbb{R}, x \in (-1, 1)$$

Illustration 16 : $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$

Solution : $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} \Rightarrow \lim_{x \rightarrow 0} \frac{1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots\right) - 2x}{x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)}$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2 \cdot \frac{x^3}{6} + 2 \cdot \frac{x^5}{5!} + \dots}{\frac{x^3}{6} + \frac{x^5}{5!} - \dots} \Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \left(\frac{1}{3} + \frac{1}{60} x^2 + \dots\right)}{x^3 \left(\frac{1}{6} + \frac{1}{120} x^2 + \dots\right)} = \frac{1/3}{1/6} = 2$$

Do yourself - 9 :

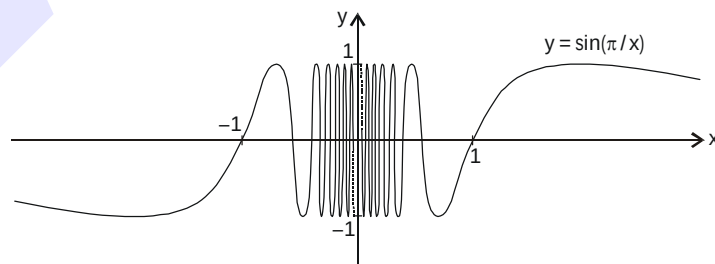
1. Evaluate : $\lim_{x \rightarrow 0} \frac{x - \sin x}{\sin(x^3)}$
2. Evaluate : $\lim_{x \rightarrow 0} \frac{x - \tan^{-1} x}{x^3}$
3. Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$
4. Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$
5. Evaluate $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \log(1 - x)}{x^3}$
6. Evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$
7. The value of $\lim_{h \rightarrow 0} \frac{\log_e(1 + 2h) - 2 \log_e(1 + h)}{h^2}$ is
(A) 0 (B) -1 (C) 1 (D) None of these
8. The value of $\lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^3}$ is
(A) 1 (B) 0 (C) 1/3 (D) 2/3
9. Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \log(1 + x)}$
10. Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$ is equal to
(A) 1/3 (B) 1/6 (C) -1/3 (D) -1/6
11. Prove using series expansion $\lim_{x \rightarrow 0} \frac{(1 + x)^n - 1}{x} = n$

Miscellaneous Illustrations :

Illustration 17: Evaluate $\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$.

Solution : Again the function $f(x) = \sin(\pi/x)$ is undefined at 0. Evaluating the function for some small values of x , we get $f(1) = \sin \pi = 0$, $f\left(\frac{1}{2}\right) = \sin 2\pi = 0$,
 $f(0.1) = \sin 10\pi = 0$, $f(0.01) = \sin 100\pi = 0$.

On the basis of this information we might be tempted to guess that $\lim_{x \rightarrow 0} \sin \frac{\pi}{x} = 0$ but this time our guess is wrong. Note that although $f(1/n) = \sin n\pi = 0$ for any integer n , it is also true that $f(x) = 1$ for infinitely many values of x that approach 0. [In fact, $\sin(\pi/x) = 1$ when $\frac{\pi}{x} = \frac{\pi}{2} + 2n\pi$ and solving for x , we get $x = 2/(4n + 1)$]. The graph of f is given in following figure



The dashed line indicate that the values of $\sin(\pi/x)$ oscillate between 1 and -1 infinitely often as x approaches 0. Since the values of $f(x)$ do not approach a fixed number as x approaches 0,

$$\Rightarrow \lim_{x \rightarrow 0} \sin \frac{\pi}{x} \text{ does not exist.}$$

Solution :

Since $f(x)$ is continuous at $x = 0$, so at $x = 0$, both left and right limits must exist and both must be equal to 3.

$$\text{Now } \lim_{x \rightarrow 0^-} \frac{a(1 - x \sin x) + b \cos x + 5}{x^2} = \lim_{x \rightarrow 0^-} \frac{(a + b + 5) + \left(-a - \frac{b}{2}\right)x^2 + \dots}{x^2} = 3$$

(By the expansions of $\sin x$ and $\cos x$)

$$\text{If } \lim_{x \rightarrow 0^-} f(x) \text{ exists then } a + b + 5 = 0 \text{ and } -a - \frac{b}{2} = 3 \Rightarrow a = -1 \text{ and } b = -4$$

$$\text{since } \lim_{x \rightarrow 0^+} \left(1 + \left(\frac{cx + dx^3}{x^2}\right)\right)^{\frac{1}{x}} \text{ exists} \Rightarrow \lim_{x \rightarrow 0^+} \frac{cx + dx^3}{x^2} = 0 \Rightarrow c = 0$$

$$\text{Now } \lim_{x \rightarrow 0^+} (1 + dx)^{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \left[(1 + dx)^{\frac{1}{dx}}\right]^d = e^d$$

$$\text{So } e^d = 3 \Rightarrow d = \ln 3,$$

$$\text{Hence } a = -1, b = -4, c = 0 \text{ and } d = \ln 3.$$

Do yourself - 1 :

1. If $f(x) = \begin{cases} \cos x; & x \geq 0 \\ x + k; & x < 0 \end{cases}$ find the value of k if $f(x)$ is continuous at $x = 0$.

2. If $f(x) = \begin{cases} \frac{|x+2|}{\tan^{-1}(x+2)} & ; \quad x \neq -2 \\ 2 & ; \quad x = -2 \end{cases}$ then discuss the continuity of $f(x)$ at $x = -2$

3. A function $y = f(x)$ is defined as $f(x) = \begin{cases} k \sin \frac{(x+3)\pi}{6} & \text{for } x \leq 2 \\ \frac{3 - \sqrt{11-x}}{x-2} & \text{for } x > 2 \end{cases}$. If $f(x)$ is continuous at

$x = 2$, then find the value of k .

4. Discuss the continuity of the function $[\cos x]$ at $x = \frac{\pi}{2}$, where $[.]$ denotes the greatest integer function

5. Find whether $f(x)$ is continuous or not at $x = 1$, if $f(x) = \begin{cases} \sin \frac{\pi x}{2} & , \quad x < 1 \\ [x] & , \quad x \geq 1 \end{cases}$

where $[.]$ denotes the greatest integer function

6. Let $f(x) = \begin{cases} x^2 & \text{if } x > 2 \\ x+1 & \text{if } x \leq 2 \end{cases}$. Discuss continuity of $f(x)$.

7. Let $f(x) = \frac{\ln(1+ax) - \ln(1-bx)}{x}$. Find the value which should be assigned to f at $x = 0$, so that it is continuous at $x = 0$.

2. CONTINUITY OF THE FUNCTION IN AN INTERVAL :

- (a) A function is said to be continuous in (a, b) if f is continuous at each & every point belonging to (a, b) .
- (b) A function is said to be continuous in a closed interval $[a, b]$ if :
- f is continuous in the open interval (a, b)
 - f is right continuous at 'a' i.e. $\lim_{x \rightarrow a^+} f(x) = f(a) = \text{a finite quantity}$
 - f is left continuous at 'b' i.e. $\lim_{x \rightarrow b^-} f(x) = f(b) = \text{a finite quantity}$

Note :

- All polynomials, trigonometrical functions, exponential & logarithmic functions are continuous in their domains.
- If $f(x)$ & $g(x)$ are two functions that are continuous at $x = c$ then the function defined by :
 $F_1(x) = f(x) \pm g(x)$; $F_2(x) = K f(x)$, where K is any real number ; $F_3(x) = f(x) \cdot g(x)$ are also continuous at $x = c$.

Further, if $g(c)$ is not zero, then $F_4(x) = \frac{f(x)}{g(x)}$ is also continuous at $x = c$.

Illustration 3 : Discuss the continuity of $f(x) = \begin{cases} |x+1|, & x < -2 \\ 2x+3, & -2 \leq x < 0 \\ x^2+3, & 0 \leq x < 3 \\ x^3-15, & x \geq 3 \end{cases}$

Solution : We write $f(x)$ as $f(x) = \begin{cases} -x-1, & x < -2 \\ 2x+3, & -2 \leq x < 0 \\ x^2+3, & 0 \leq x < 3 \\ x^3-15, & x \geq 3 \end{cases}$

As we can see, $f(x)$ is defined as a polynomial function in each of intervals $(-\infty, -2)$, $(-2, 0)$, $(0, 3)$ and $(3, \infty)$. Therefore, it is continuous in each of these four open intervals. Thus we check the continuity at $x = -2, 0, 3$.

At the point $x = -2$

$$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} (-x - 1) = +2 - 1 = 1$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (2x + 3) = 2 \cdot (-2) + 3 = -1$$

Therefore, $\lim_{x \rightarrow -2} f(x)$ does not exist and hence $f(x)$ is discontinuous at $x = -2$.

At the point $x = 0$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (2x + 3) = 3$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^2 + 3) = 3$$

$$f(0) = 0^2 + 3 = 3$$

Therefore $f(x)$ is continuous at $x = 0$.

At the point $x = 3$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x^2 + 3) = 3^2 + 3 = 12$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (x^3 - 15) = 3^3 - 15 = 12$$

$$f(3) = 3^3 - 15 = 12$$

Therefore, $f(x)$ is continuous at $x = 3$.

We find that $f(x)$ is continuous at all points in $\mathbb{R}$ except at $x = -2$

Do yourself -2 :

1. If $f(x) = \begin{cases} \frac{x^2}{a} & ; 0 \leq x < 1 \\ -1 & ; 1 \leq x < \sqrt{2} \\ \frac{2b^2 - 4b}{x^2} & ; \sqrt{2} \leq x < \infty \end{cases}$ then find the value of a & b if $f(x)$ is continuous in $[0, \infty)$

2. Discuss the continuity of $f(x) = \begin{cases} |x-3| & ; 0 \leq x < 1 \\ \sin x & ; 1 \leq x \leq \frac{\pi}{2} \\ \log_{\frac{\pi}{2}} x & ; \frac{\pi}{2} < x < 3 \end{cases}$ in $[0, 3)$

3. Examine the continuity of the function $f(x) = \begin{cases} \frac{x^2 - 1}{x^2 - 2|x-1| - 1} & , x \neq 1 \\ \frac{1}{2} & , x = 1 \end{cases}$ at $x = 1$.

4. Show that the function $f(x) = \begin{cases} 2x + 3, & -3 \leq x < -2 \\ x + 1, & -2 \leq x < 0 \\ x + 2, & 0 \leq x \leq 1 \end{cases}$ is discontinuous at $x = 0$ and continuous at other points in the interval $[-3, 1]$

5. Let $f(x) = \begin{cases} -2 \sin x & \text{if } x \leq -\pi/2 \\ A \sin x + B & \text{if } -\frac{\pi}{2} < x < \pi/2 \\ \cos x & \text{if } x \geq \pi/2 \end{cases}$ Find A and B so as to make the function continuous.

6. Find the intervals on which f and g are continuous where $f(x) = \begin{cases} 3-x & \text{if } -5 \leq x < 2 \\ x-2 & \text{if } 2 \leq x < 5 \end{cases}$ and

$$g(x) = \begin{cases} 2-x & \text{if } -5 \leq x < 2 \\ x-2 & \text{if } 2 \leq x < 5 \end{cases}$$

7. Discuss continuity of the function $f(x) = \frac{2x^5 - 8x^2 + 11}{x^4 + 4x^3 + 8x^2 + 8x + 4}$

3. THE INTERMEDIATE VALUE THEOREM :

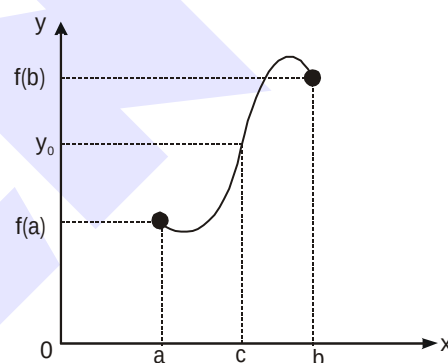
Suppose $f(x)$ is continuous on an interval

I, and a and b are any two points of I.

Then if y_0 is a number between $f(a)$ and

$f(b)$, there exists a number c between a

and b such that $f(c) = y_0$



The function f, being continuous on $[a, b]$ takes on every value between $f(a)$ and $f(b)$

Note that a function f which is continuous in $[a, b]$ possesses the following properties :

- (i) If $f(a)$ & $f(b)$ possess opposite signs, then there exists at least one root of the equation $f(x) = 0$ in the open interval (a, b) .
- (ii) If K is any real number between $f(a)$ & $f(b)$, then there exists at least one root of the equation $f(x) = K$ in the open interval (a, b) .

Note : In above cases the number of roots is always odd.

Illustration 4 : Show that the function, $f(x) = (x-a)^2(x-b)^2 + x$, takes the value $\frac{a+b}{2}$ for some

$$x_0 \in (a, b)$$

Solution :

$$f(x) = (x-a)^2(x-b)^2 + x$$

$$f(a) = a$$

$$f(b) = b$$

$$\& \frac{a+b}{2} \in (f(a), f(b))$$

$\therefore$ By intermediate value theorem, there is at least one $x_0 \in (a, b)$ such that $f(x_0) = \frac{a+b}{2}$.

Illustration 5 : Let $f : [0, 1] \xrightarrow{\text{onto}} [0, 1]$ be a continuous function, then prove that $f(x) = x$ for atleast one $x \in [0, 1]$

Solution : Consider $g(x) = f(x) - x$

$$g(0) = f(0) - 0 = f(0) \geq 0 \quad \left\{ \because 0 \leq f(x) \leq 1 \right\}$$

$$g(1) = f(1) - 1 \leq 0$$

$$\Rightarrow g(0) \cdot g(1) \leq 0$$

$$\Rightarrow g(x) = 0 \text{ has atleast one root in } [0, 1]$$

$$\Rightarrow f(x) = x \text{ for atleast one } x \in [0, 1]$$

Do yourself -3 :

1. If $f(x)$ is continuous in $[a, b]$ then prove that $\exists c \in [a, b]$ such that $f(c) = \frac{2f(a) + 3f(b)}{5}$.
2. Show that the function $f(x) = x - 1 - \cos \pi x$, $x \in [1, 2]$ takes on the value $1/2$.
3. Does the function $f(x) = x^3/4 - \sin \pi x + 3$ take on the value $2\frac{1}{3}$ within the interval $[-2, 2]$?
4. Show that the function $f(x) = (x-a)^2(x-b)^2 + x$ take the value $\frac{a+b}{2}$ for some value of $x \in [a, b]$.
5. Use intermediate value theorem to show that the equation $2x^3 + x^2 - x + 1 = 5$ has a solution in the interval $[1, 2]$.

4. SOME IMPORTANT POINTS :

- (a) If $f(x)$ is continuous & $g(x)$ is discontinuous at $x = a$ then the product function $\phi(x) = f(x) \cdot g(x)$ will **not necessarily be discontinuous at $x = a$** , e.g.

$$f(x) = x \text{ \& } g(x) = \begin{cases} \sin \frac{\pi}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$f(x)$ is continuous at $x = 0$ & $g(x)$ is discontinuous at $x = 0$, but $f(x) \cdot g(x)$ is continuous at $x = 0$.

- (b) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then the product function $\phi(x) = f(x) \cdot g(x)$ **is not necessarily be discontinuous at $x = a$** , e.g.

$$f(x) = -g(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

$f(x)$ & $g(x)$ both are discontinuous at $x = 0$ but the product function $f \cdot g(x)$ is still continuous at $x = 0$

- (c) If $f(x)$ and $g(x)$ both are discontinuous at $x = a$ then $f(x) \pm g(x)$ is not necessarily be discontinuous at $x = a$

- (d) A continuous function whose domain is closed must have a range also in closed interval.
- (e) If f is continuous at $x = a$ & g is continuous at $x = f(a)$ then the composite $g[f(x)]$ is continuous at $x = a$.

eg. $f(x) = \frac{x \sin x}{x^2 + 2}$ & $g(x) = |x|$ are continuous at $x = 0$, hence the composite $(g \circ f)(x) = \left| \frac{x \sin x}{x^2 + 2} \right|$

will also be continuous at $x = 0$

Illustration 6: If $f(x) = \frac{x+1}{x-1}$ and $g(x) = \frac{1}{x-2}$, then discuss the continuity of $f(x)$, $g(x)$ and $f \circ g(x)$ in its domain.

Solution : $f(x) = \frac{x+1}{x-1}$

$f(x)$ is a rational function it must be continuous in its domain and f is not defined at $x = 1$.

$$g(x) = \frac{1}{x-2}$$

$g(x)$ is also a rational function. It must be continuous in its domain and g is not defined at $x = 2$.

Consider $g(x) = 1$

$$\frac{1}{x-2} = 1 \Rightarrow x = 3$$

$\therefore f \circ g(x)$ is continuous in its domain : $\mathbb{R} - \{2, 3\}$

Do yourself -4 :

- Let $f(x) = [x]$ & $g(x) = \text{sgn}(x)$ (where $[.]$ denotes greatest integer function), then discuss the continuity of $f(x) \pm g(x)$, $f(x) \cdot g(x)$ & $\frac{f(x)}{g(x)}$ at $x = 0$.
- If $f(x) = \sin|x|$ & $g(x) = \tan|x|$ then discuss the continuity of $f(x) \pm g(x)$; $\frac{f(x)}{g(x)}$ & $f(x) \cdot g(x)$
- Discuss the continuity of $f(x) = [x] + |x-1|$. (where $[.]$ denotes greatest integer function)
- If $f(x) = [\sin(x-1)] - \{\sin(x-1)\}$ then comment on continuity of $f(x)$ at $x = \frac{\pi}{2} + 1$.
(where $[.]$ & $\{.\}$ denote greatest integer function & fractional part function)

5. CONTINUITY OVER COUNTABLE SET :

There are functions which are continuous over a countable set and else where discontinuous.

Illustration 7: If $f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ -x & \text{if } x \notin \mathbb{Q} \end{cases}$, find the points where $f(x)$ is continuous

Solution : Let $x = a$ be the point at which $f(x)$ is continuous.

$$\Rightarrow \lim_{\substack{x \rightarrow a \\ \text{through rational}}} f(x) = \lim_{\substack{x \rightarrow a \\ \text{through irrational}}} f(x)$$

$$\Rightarrow a = -a$$

$$\Rightarrow a = 0 \Rightarrow \text{function is continuous at } x = 0.$$

Do yourself -5 :

1. If $g(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$, then find the points where function is continuous.
2. If $f(x) = \begin{cases} x^2 & ; x \in \mathbb{Q} \\ 1-x^2 & ; x \notin \mathbb{Q} \end{cases}$, then find the points where function is continuous.
3. Let $f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ -1 & \text{if } x \text{ is irrational} \end{cases}$. Discuss continuity of the function $f(x)$.
4. Let $f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ 5x, & \text{if } x \text{ is irrational} \end{cases}$. Discuss continuity of the function $f(x)$.

DIFFERENTIABILITY

1. MEANING OF DERIVATIVE :

The instantaneous rate of change of a function with respect to the dependent variable is called derivative. Let 'f' be a given function of one variable and let Δx denote a number (positive or negative) to be added to the number x . Let Δf denote the corresponding change of 'f' then $\Delta f = f(x + \Delta x) - f(x)$

$$\Rightarrow \frac{\Delta f}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

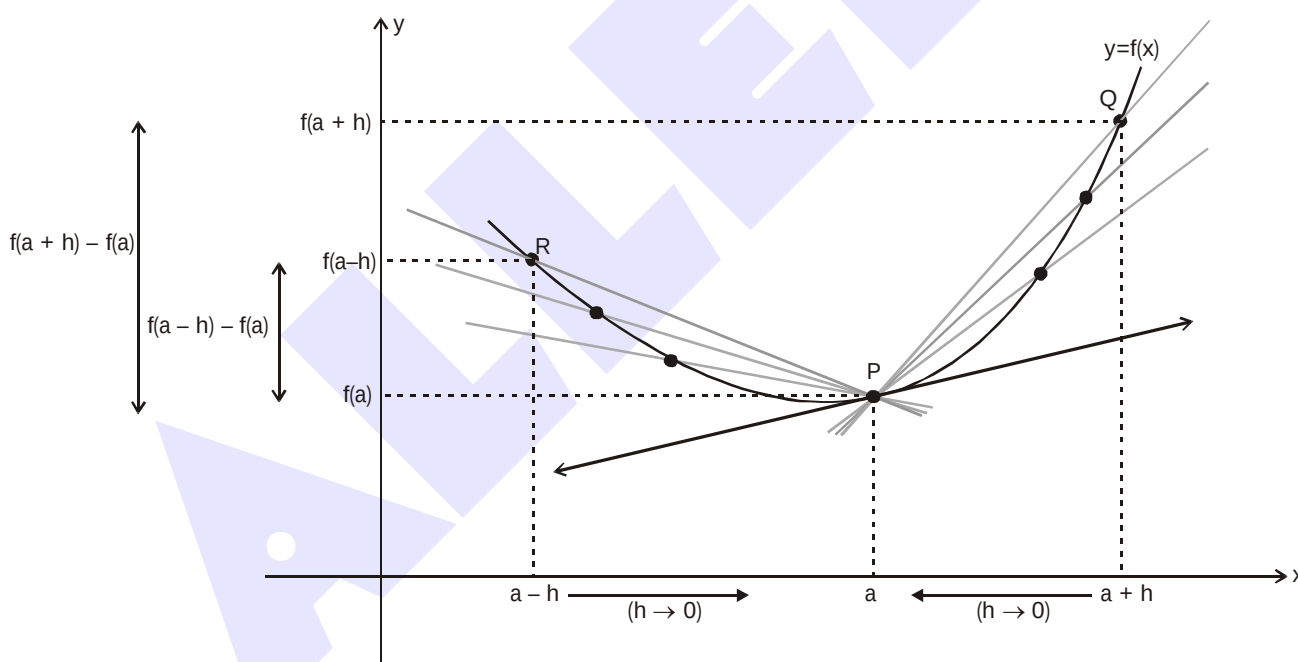
If $\Delta f/\Delta x$ approaches a limit as Δx approaches zero, this limit is the derivative of 'f' at the point x . The derivative of a function 'f' is a function ; this function is denoted by symbols such as

$$f'(x), \frac{df}{dx}, \frac{d}{dx}f(x) \text{ or } \frac{df(x)}{dx}$$

$$\Rightarrow \frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

The derivative evaluated at a point a , is written, $f'(a), \left. \frac{df(x)}{dx} \right|_{x=a}, f'(x)_{x=a}$, etc.

2. EXISTENCE OF DERIVATIVE AT $x = a$:



(a) Right hand derivative :

The right hand derivative of $f(x)$ at $x = a$ denoted by $Rf'(a)$ is defined as :

$$Rf'(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}, \text{ provided the limit exists \& is finite. } (h > 0)$$

(b) Left hand derivative :

The left hand derivative of $f(x)$ at $x = a$ denoted by $Lf'(a)$ is defined as :

$$Lf'(a) = \lim_{h \rightarrow 0^+} \frac{f(a-h) - f(a)}{-h}, \text{ provided the limit exists \& is finite. } (h > 0)$$

Hence $f(x)$ is said to be **derivable or differentiable at $x = a$** . If $Rf'(a) = Lf'(a) = \text{finite quantity}$

and it is denoted by $f'(a)$; where $f'(a) = Lf'(a) = Rf'(a)$ & it is called derivative or differential coefficient of $f(x)$ at $x = a$.

3. DIFFERENTIABILITY & CONTINUITY :

Theorem : If a function $f(x)$ is derivable at $x = a$, then $f(x)$ is continuous at $x = a$.

Proof : $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists.

$$\text{Also } f(a+h) - f(a) = \frac{f(a+h) - f(a)}{h} \cdot h \quad [h \neq 0]$$

$$\therefore \lim_{h \rightarrow 0} [f(a+h) - f(a)] = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \cdot h = f'(a) \cdot 0 = 0$$

$$\Rightarrow \lim_{h \rightarrow 0} [f(a+h) - f(a)] = 0 \Rightarrow \lim_{h \rightarrow 0} f(a+h) = f(a) \Rightarrow f(x) \text{ is continuous at } x = a.$$

Note :

(i) Differentiable $\Rightarrow$ Continuous ; Continuity $\nRightarrow$ Differentiable ; Not Differentiable $\nRightarrow$ Not Continuous
But Not Continuous $\Rightarrow$ Not Differentiable

(ii) All polynomial, rational, trigonometric, logarithmic and exponential function are continuous and differentiable in their domains.

(iii) If $f(x)$ & $g(x)$ are differentiable at $x = a$, then the functions $f(x) + g(x)$, $f(x) - g(x)$, $f(x) \cdot g(x)$ will also be differentiable at $x = a$ & if $g(a) \neq 0$ then the function $f(x)/g(x)$ will also be differentiable at $x = a$.

Illustration 1 : Let $f(x) = \begin{cases} \operatorname{sgn}(x) + x; & -\infty < x < 0 \\ -1 + \sin x; & 0 \leq x < \frac{\pi}{2} \\ \cos x; & \frac{\pi}{2} \leq x < \infty \end{cases}$. Discuss the continuity & differentiability at

$x = 0$ & $\frac{\pi}{2}$.

Solution : $f(x) = \begin{cases} -1 + x; & -\infty < x < 0 \\ -1 + \sin x; & 0 \leq x < \frac{\pi}{2} \\ \cos x; & \frac{\pi}{2} \leq x < \infty \end{cases}$

To check the differentiability at $x = 0$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{-1 + 0 - h - (-1)}{-h} = 1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{-1 + \sinh h + 1}{h} = 1$$

$$\therefore \text{LHD} = \text{RHD}$$

$$\therefore \text{Differentiable at } x = 0.$$

$$\Rightarrow \text{Continuous at } x = 0.$$

To check the continuity at $x = \frac{\pi}{2}$

$$\text{LHL } \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} (-1 + \sin x) = 0$$

$$\text{RHL } \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^+} \cos x = 0$$

$$\therefore \text{LHL} = \text{RHL} = f\left(\frac{\pi}{2}\right) = 0$$

$$\therefore \text{Continuous at } x = \frac{\pi}{2}.$$

To check the differentiability at $x = \frac{\pi}{2}$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2} - h\right) - f\left(\frac{\pi}{2}\right)}{-h} = \lim_{h \rightarrow 0} \frac{-1 + \cosh h - 0}{-h} = 0$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f\left(\frac{\pi}{2} + h\right) - f\left(\frac{\pi}{2}\right)}{h} = \lim_{h \rightarrow 0} \frac{-\sinh h - 0}{h} = -1$$

$$\therefore \text{LHD} \neq \text{RHD}$$

$$\therefore \text{not differentiable at } x = \frac{\pi}{2}.$$

Illustration 2 : If $f(x) = \begin{cases} A + Bx^2 & x < 1 \\ 3Ax - B + 2 & x \geq 1 \end{cases}$

then find A and B so that f(x) become differentiable at $x = 1$.

Solution : $\text{Rf}'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{3A(1+h) - B + 2 - 3A + B - 2}{h} = \lim_{h \rightarrow 0} \frac{3Ah}{h} = 3A$

$$\text{Lf}'(1) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{A + B(1-h)^2 - 3A + B - 2}{-h} = \lim_{h \rightarrow 0} \frac{(-2A + 2B - 2) + Bh^2 - 2Bh}{-h}$$

hence for this limit to be defined

$$-2A + 2B - 2 = 0$$

$$B = A + 1$$

$$\text{Lf}'(1) = \lim_{h \rightarrow 0} -(Bh - 2B) = 2B$$

$$\therefore \text{Lf}'(1) = \text{Rf}'(1)$$

$$3A = 2B = 2(A + 1)$$

$$A = 2, B = 3$$

Illustration 3 : $f(x) = \begin{cases} [\cos \pi x] & x \leq 1 \\ 2\{x\} - 1 & x > 1 \end{cases}$ comment on the derivability at $x=1$, where $[]$ denotes greatest

integer function & $\{ \}$ denotes fractional part function.

Solution : $Lf'(1) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{[\cos(\pi - \pi h)] + 1}{-h} = \lim_{h \rightarrow 0} \frac{-1 + 1}{-h} = 0$

$$Rf'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{2\{1+h\} - 1 + 1}{h} = \lim_{h \rightarrow 0} \frac{2h}{h} = 2$$

Hence $f(x)$ is not differentiable at $x = 1$.

Do yourself -1 :

1. A function is defined as follows :

$$f(x) = \begin{cases} x^3 & ; \quad x^2 < 1 \\ x & ; \quad x^2 \geq 1 \end{cases} \text{ discuss the continuity and differentiability at } x = 1.$$

2. If $f(x) = \begin{cases} ax^3 + b, & \text{for } 0 \leq x \leq 1 \\ 2 \cos \pi x + \tan^{-1} x, & \text{for } 1 < x \leq 2 \end{cases}$ be the differentiable function in $[0, 2]$, then find a

and b. (where $[.]$ denotes the greatest integer function)

3. Check whether derivative exists at indicated points

(a) $f(x) = (x+1)^{1/3}$ at $x = -1$

(b) $f(x) = 3|x-1|$ at $x = 1$

(c) $f(x) = [x+1]$ at $x = 0$

(where $[.]$ denotes the greatest integer function)

4. If $f(x) = \begin{cases} \ln x & \text{for } x \geq e^2 \\ ax + b & \text{for } x < e^2 \end{cases}$ is differentiable on $\mathbb{R}$ then find value of a and b.

5. If $f(x) = \begin{cases} x & , \quad x \leq 1 \\ x^2 + bx + c & , \quad x > 1 \end{cases}$ then find the values of b and c if $f(x)$ is differentiable at $x = 1$

6. Discuss differentiability of $f(x) = \begin{cases} x^3 - 1 & ; \quad 1 < x < \infty \\ x - 1 & ; \quad -\infty < x \leq 1 \end{cases}$ at $x = 1$

4. IMPORTANT NOTE :

(a) Let $Rf'(a) = p$ & $Lf'(a) = q$ where p & q are finite then :

- (i) $p = q \Rightarrow f$ is differentiable at $x = a \Rightarrow f$ is continuous at $x = a$
 (ii) $p \neq q \Rightarrow f$ is not differentiable at $x = a$, but f is continuous at $x = a$.

Illustration 4 : Determine the values of x for which the following functions fails to be continuous or

$$\text{differentiable } f(x) = \begin{cases} (1-x), & x < 1 \\ (1-x)(2-x), & 1 \leq x \leq 2 \\ (3-x), & x > 2 \end{cases}, \text{ Justify your answer.}$$

Solution :

By the given definition it is clear that the function f is continuous and differentiable at all points except possibly at $x = 1$ and $x = 2$.

Check the differentiability at $x = 1$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{1 - (1-h) - 0}{-h} = -1$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{\{1 - (1+h)\} \{2 - (1+h)\} - 0}{h} = -1$$

$\therefore q = p \therefore$ Differentiable at $x = 1. \Rightarrow$ Continuous at $x = 1$.

Check the differentiability at $x = 2$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} = \lim_{h \rightarrow 0} \frac{(1-2+h)(2-2+h) - 0}{-h} = 1 = \text{finite}$$

$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{(3-2-h) - 0}{h} \rightarrow \infty \text{ (not finite)}$$

$\therefore q \neq p \therefore$ not differentiable at $x = 2$.

Now we have to check the continuity at $x = 2$

$$\text{LHL} = \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (1-x)(2-x) = \lim_{h \rightarrow 0} (1-(2-h))(2-(2-h)) = 0$$

$$\text{RHL} = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (3-x) = \lim_{h \rightarrow 0} (3-(2+h)) = 1$$

$\therefore \text{LHL} \neq \text{RHL}$

$\Rightarrow$ not continuous at $x = 2$.

Do yourself - 2 :

1. Let $f(x) = (x-1)|x-1|$. Discuss the continuity and differentiability of $f(x)$ at $x = 1$.

2. The function $f(x) = \begin{cases} \frac{\pi}{4} + \tan^{-1} x, & |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & |x| > 1 \end{cases}$. Discuss the continuity and differentiability of $f(x)$:

3. Which of the following functions is differentiable at $x = 0$?

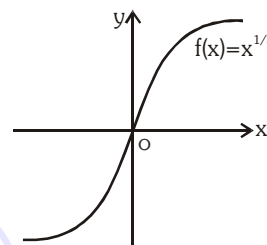
- (A) $\cos(|x|) + |x|$ (B) $\cos(|x|) - |x|$ (C) $\sin(|x|) + |x|$ (D) $\sin(|x|) - |x|$

(b) Vertical tangent :

- (i) If $y = f(x)$ is continuous at $x = a$ and $\lim_{x \rightarrow a} |f'(x)|$ approaches to ∞ , then $y = f(x)$ has a vertical tangent at $x = a$. If a function has vertical tangent at $x = a$ then it is non differentiable at $x = a$.

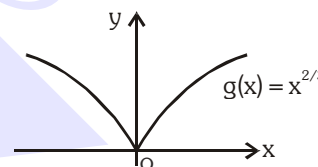
e.g. (1) $f(x) = x^{1/3}$ has vertical tangent at $x = 0$

since $f'_+(0) \rightarrow \infty$ and $f'_-(0) \rightarrow \infty$ hence $f(x)$ is not differentiable at $x = 0$



(2) $g(x) = x^{2/3}$ have vertical tangent at $x = 0$

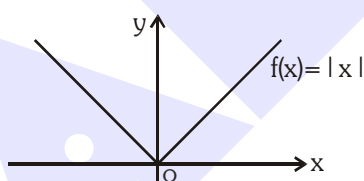
since $g'_+(0) \rightarrow \infty$ and $g'_-(0) \rightarrow -\infty$ hence $g(x)$ is not differentiable at $x = 0$.



(c) Geometrical interpretation of differentiability :

- (i) If the function $y = f(x)$ is differentiable at $x = a$, then a unique non vertical tangent can be drawn to the curve $y = f(x)$ at the point $P(a, f(a))$ & $f'(a)$ represent the slope of the tangent at point P.
- (ii) If a function $f(x)$ does not have a unique tangent (p & q are finite but unequal), then f is continuous at $x = a$, it geometrically implies a corner at $x = a$.

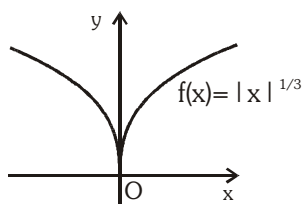
e.g. $f(x) = |x|$ is continuous but not differentiable at $x = 0$ & there is corner at $x = 0$.



(does not have unique tangent, corner at $x = 0$) $\begin{cases} p = 1 \\ q = -1 \end{cases}$

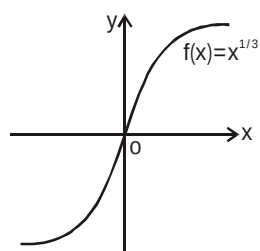
- (iii) If one of p & q tends to ∞ and other tends to $-\infty$, then there will be a cusp at $x = a$. Where $p = Rf'(a)$ and $q = Lf'(a)$

e.g. (1) $f(x) = |x|^{1/3}$ is continuous but not differentiable at $x = 0$ & there is cusp at $x = 0$.



(has a vertical tangent, cusp at $x = 0$) $\begin{cases} p \rightarrow +\infty \\ q \rightarrow -\infty \end{cases}$

- (2) $f(x) = x^{1/3}$ is continuous but not differentiable at $x = 0$ because $Rf'(0) \rightarrow \infty$ and $Lf'(0) \rightarrow \infty$.



$$\begin{cases} p \rightarrow +\infty \\ q \rightarrow +\infty \end{cases}$$

(have a unique vertical tangent but does not have corner)

Note : corner/cusp/vertical tangent $\Rightarrow$ non differentiable

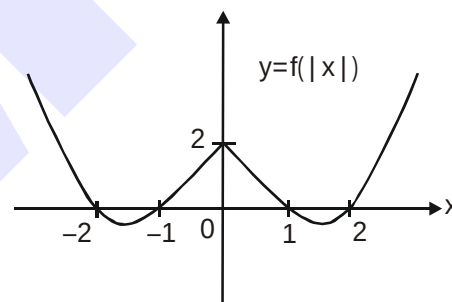
non differentiable $\nRightarrow$ corner/cusp/vertical tangent

Illustration 5 : If $f(x) = \begin{cases} x-3 & x < 0 \\ x^2 - 3x + 2 & x \geq 0 \end{cases}$. Draw the graph of the function & discuss the continuity and differentiability of $f(|x|)$ and $|f(x)|$.

Solution :

$$f(|x|) = \begin{cases} |x| - 3; & |x| < 0 \rightarrow \text{not possible} \\ |x|^2 - 3|x| + 2; & |x| \geq 0 \end{cases}$$

$$f(|x|) = \begin{cases} x^2 + 3x + 2, & x < 0 \\ x^2 - 3x + 2, & x \geq 0 \end{cases}$$



at $x = 0$

$$q = \text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{-h} = 3$$

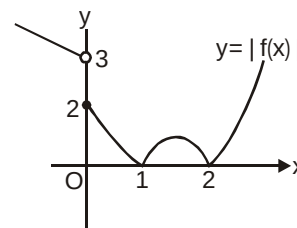
$$p = \text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{h} = -3$$

$$\therefore q \neq p$$

$\therefore$ not differentiable at $x = 0$. but p & q are both are finite

$\Rightarrow$ continuous at $x = 0$

$$\text{Now, } |f(x)| = \begin{cases} 3-x, & x < 0 \\ (x^2 - 3x + 2), & 0 \leq x < 1 \\ -(x^2 - 3x + 2), & 1 \leq x \leq 2 \\ (x^2 - 3x + 2), & x > 2 \end{cases}$$



To check differentiability at $x = 0$

$$\left. \begin{aligned} q = \text{LHD} &= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{3+h-2}{-h} = \lim_{h \rightarrow 0} \frac{(1+h)}{-h} \rightarrow -\infty \\ p = \text{RHD} &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 - 3h + 2 - 2}{h} = -3 \end{aligned} \right\} \Rightarrow \text{not differentiable at } x = 0.$$

Now to check continuity at $x = 0$

$$\left. \begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 3 - x = 3 \\ \text{RHL} &= \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x^2 - 3x + 2 = 2 \end{aligned} \right\} \Rightarrow \text{not continuous at } x = 0.$$

To check differentiability at $x = 1$

$$\begin{aligned} q = \text{LHD} &= \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{(1-h)^2 - 3(1-h) + 2 - 0}{-h} = \lim_{h \rightarrow 0} \frac{h^2 + h}{-h} = -1 \\ p = \text{RHD} &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{-(h^2 + 2h + 1 - 3 + 3h + 2) - 0}{h} = \lim_{h \rightarrow 0} \frac{-(h^2 - h)}{h} = 1 \\ \Rightarrow &\text{not differentiable at } x = 1. \end{aligned}$$

but $|f(x)|$ is continuous at $x = 1$, because $p \neq q$ and both are finite.

To check differentiability at $x = 2$

$$\begin{aligned} q = \text{LHD} &= \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{-(4 + h^2 - 4h - 6 + 3h + 2) - 0}{-h} = \lim_{h \rightarrow 0} \frac{h^2 - h}{h} = -1 \\ p = \text{RHD} &= \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{(h^2 + 4h + 4 - 6 - 3h + 2) - 0}{h} = \lim_{h \rightarrow 0} \frac{(h^2 + h)}{h} = 1 \\ \Rightarrow &\text{not differentiable at } x = 2. \end{aligned}$$

but $|f(x)|$ is continuous at $x = 2$, because $p \neq q$ and both are finite.

Do yourself - 3 :

1. Let $f(x) = \begin{cases} -4 & ; -4 < x < 0 \\ x^2 - 4 & ; 0 \leq x < 4 \end{cases}$

Discuss the continuity and differentiability of $g(x) = |f(x)|$.

- Let $f(x) = \min \{|x-1|, |x+1|, 1\}$. Find the number of points where it is not differentiable.
- Consider function $f(x) = \min \{\tan x, \cot x\}$ in $(0, 2\pi)$. Number of points where f is either fails to be derivable is 'm' and number of points where it is discontinuous is 'n'. Find (m, n)
- Let $f(x) = \max \{2\sin x, 1 - \cos x\} \forall x \in (0, \pi)$. Discuss differentiability of $f(x)$
- Let $f(x) = \max \{1 - x, 1 + x, 2\}$. Discuss differentiability of $f(x)$

5. DIFFERENTIABILITY OVER AN INTERVAL :

- $f(x)$ is said to be differentiable over an open interval (a, b) if it is differentiable at each & every point of the open interval (a, b) .
- $f(x)$ is said to be differentiable over the closed interval $[a, b]$ if :
 - $f(x)$ is differentiable in (a, b) &
 - for the points a and b , $f'(a^+)$ & $f'(b^-)$ exist.

Illustration 6 : If $f(x) = \begin{cases} e^{-|x|}, & -5 < x < 0 \\ -e^{-|x-1|} + e^{-1} + 1, & 0 \leq x < 2 \\ e^{-|x-2|}, & 2 \leq x < 4 \end{cases}$

Discuss the continuity and differentiability of $f(x)$ in the interval $(-5, 4)$.

Solution : $f(x) = \begin{cases} e^{+x} & -5 < x < 0 \\ -e^{x-1} + e^{-1} + 1 & 0 \leq x \leq 1 \\ -e^{-x+1} + e^{-1} + 1 & 1 < x < 2 \\ e^{-x+2} & 2 \leq x < 4 \end{cases}$

Check the differentiability at $x = 0$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{e^{-h} - 1}{-h} = 1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{-e^{h-1} + e^{-1} + 1 - 1}{h} = -e^{-1}$$

$\therefore \text{LHD} \neq \text{RHD}$

$\therefore$ Not differentiable at $x = 0$, but continuous at $x = 0$ since LHD and RHD both are finite.

Check the differentiability at $x = 1$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{-e^{1-h-1} + e^{-1} + 1 - e^{-1}}{-h} = -1$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{-e^{1-h-1} + e^{-1} + 1 - e^{-1}}{h} = 1$$

$\therefore \text{LHD} \neq \text{RHD}$

$\therefore$ Not differentiable at $x = 1$, but continuous at $x = 1$ since LHD and RHD both are finite.

Check the differentiability at $x = 2$

$$\text{LHD} = \lim_{h \rightarrow 0} \frac{f(2-h) - f(2)}{-h} = \lim_{h \rightarrow 0} \frac{-e^{-2+h+1} + e^{-1} + 1 - 1}{-h} = \lim_{h \rightarrow 0} \frac{-e^{-1}(e^h - 1)}{-h} = e^{-1}$$

$$\text{RHD} = \lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0} \frac{e^{-h} - 1}{h} = -1$$

$\therefore \text{LHD} \neq \text{RHD}$

$\therefore$ Not differentiable at $x = 2$, but continuous at $x = 2$ since LHD & RHD both are finite.

Note :

- (i) If $f(x)$ is differentiable at $x = a$ & $g(x)$ is not differentiable at $x = a$, then the product function $F(x) = f(x).g(x)$ can still be differentiable at $x = a$.

e.g. Consider $f(x) = x$ & $g(x) = |x|$. f is differentiable at $x = 0$ & g is non-differentiable at $x = 0$, but $f(x).g(x)$ is still differentiable at $x = 0$.

- (ii) If $f(x)$ & $g(x)$ both are not differentiable at $x = a$ then the product function; $F(x) = f(x).g(x)$ can still be differentiable at $x = a$.

e.g. Consider $f(x) = |x|$ & $g(x) = -|x|$. f & g are both non differentiable at $x = 0$, but $f(x).g(x)$ still differentiable at $x = 0$.

- (iii) If $f(x)$ & $g(x)$ both are non-differentiable at $x = a$ then the sum function $F(x) = f(x) + g(x)$ may be a differentiable function.

e.g. $f(x) = |x|$ & $g(x) = -|x|$. f & g are both non differentiable at $x = 0$, but $(f+g)(x)$ still differentiable at $x = 0$.

- (iv) If $f(x)$ is differentiable at $x = a \Rightarrow f'(x)$ is continuous at $x = a$.

$$\text{e.g. } f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Do yourself - 4 :

1. Let $f(x) = \max \{ \sin x, 1/2 \}$, where $0 \leq x \leq \frac{5\pi}{2}$. Find the number of points where it is not differentiable.

2. Let $f(x) = \begin{cases} [x] & ; 0 < x \leq 2 \\ 2x - 2 & ; 2 < x < 3 \end{cases}$, where $[.]$ denotes greatest integer function.

(a) Find that points at which continuity and differentiability should be checked.

(b) Discuss the continuity & differentiability of $f(x)$ in the interval $(0, 3)$.

3. Let $f(x) = \begin{cases} |1 - 4x^2| & , 0 \leq x < 1 \\ [x^2 - 2x] & , 1 \leq x < 2 \end{cases}$ check the differentiability in $(0, 2)$ where $[]$ denotes greatest integer function.

4. Discuss the differentiability of $f(x) = [x]x$ in $x \in (-1, 2]$ where $[]$ denotes greatest integer function.

5. Let $f(x) = \begin{cases} 1 & , -\infty < x < 0 \\ 1 + |\sin x| & , 0 \leq x < \frac{\pi}{2} \\ 2 + \left(x - \frac{\pi}{2}\right)^2 & , \frac{\pi}{2} \leq x < \infty \end{cases}$.

Discuss continuity and differentiability at $x = 0$ and $x = \frac{\pi}{2}$

6. DETERMINATION OF FUNCTION WHICH SATISFYING THE GIVEN FUNCTIONAL EQUATION :

Illustration 7 : Let $f(x+y) = f(x) + f(y) - 2xy - 1$ for all x and y . If $f'(0)$ exists and $f'(0) = -\sin\alpha$, then find $f\{f'(0)\}$.

Solution :

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\{f(x) + f(h) - 2xh - 1\} - f(x)}{h} \quad \text{(Using the given relation)}$$

$$= \lim_{h \rightarrow 0} -2x + \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = -2x + \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

[Putting $x = 0 = y$ in the given relation we find $f(0) = f(0) + f(0) + 0 - 1 \Rightarrow f(0) = 1$]

$$\therefore f'(x) = -2x + f'(0) = -2x - \sin\alpha$$

$$\Rightarrow f(x) = -x^2 - (\sin\alpha) \cdot x + c$$

$$f(0) = -0 - 0 + c \Rightarrow c = 1$$

$$\therefore f(x) = -x^2 - (\sin\alpha) \cdot x + 1$$

So, $f\{f'(0)\} = f(-\sin\alpha) = -\sin^2\alpha + \sin^2\alpha + 1$

$$\therefore f\{f'(0)\} = 1.$$

Do yourself - 5 :

1. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the equation $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$, $f(x) \neq 0$. Suppose that the function is differentiable everywhere and $f'(0) = 2$. Prove that $f'(x) = 2f(x)$.
2. Suppose f is a derivable function that satisfies the equation $f(x+y) = f(x) + f(y) + x^2y + xy^2$ for all real numbers x and y . Where $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$ then find
 - (a) $f(0)$
 - (b) $f'(0)$
 - (c) $f'(x)$
3. A function $f : \mathbb{R} \rightarrow (0, \infty)$ satisfies the equation $f(x+y) = f(x) \cdot f(y)$. If $f(x) \neq 0$ for any x , is differentiable on $\mathbb{R}$ and $f'(0) = 2$ then find $f(x)$
4. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the condition $f\left(\frac{x+y}{k}\right) = \frac{f(x)+f(y)}{k}$ ($k \neq 0, 2$). If $f(x)$ is differentiable on $\mathbb{R}$ and $f'(0) = m$ then find $f(x)$

Miscellaneous Illustrations :

Illustration 8 : Discuss the continuity and differentiability of the function $y = f(x)$ defined parametrically ;
 $x = 2t - |t - 1|$ and $y = 2t^2 + t|t|$.

Solution : Here $x = 2t - |t - 1|$ and $y = 2t^2 + t|t|$.

Now when $t < 0$;

$$x = 2t - \{-(t - 1)\} = 3t - 1 \quad \text{and} \quad y = 2t^2 - t^2 = t^2 \Rightarrow y = \frac{1}{9}(x + 1)^2$$

when $0 \leq t < 1$

$$x = 2t - (-(t - 1)) = 3t - 1 \quad \text{and} \quad y = 2t^2 + t^2 = 3t^2 \Rightarrow y = \frac{1}{3}(x + 1)^2$$

when $t \geq 1$;

$$x = 2t - (t - 1) = t + 1 \quad \text{and} \quad y = 2t^2 + t^2 = 3t^2 \Rightarrow y = 3(x - 1)^2$$

$$\text{Thus, } y = f(x) = \begin{cases} \frac{1}{9}(x + 1)^2, & x < -1 \\ \frac{1}{3}(x + 1)^2, & -1 \leq x < 2 \\ 3(x - 1)^2, & x \geq 2 \end{cases}$$

We have to check differentiability at $x = -1$ and 2 .

Differentiability at $x = -1$;

$$\text{LHD} = f'_-(-1) = \lim_{h \rightarrow 0} \frac{f(-1 - h) - f(-1)}{-h} = \lim_{h \rightarrow 0} \frac{\frac{1}{9}(-1 - h + 1)^2 - 0}{-h} = 0$$

$$\text{RHD} = f'_+(-1) = \lim_{h \rightarrow 0} \frac{f(-1 + h) - f(-1)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{3}(-1 + h + 1)^2 - 0}{h} = 0$$

Hence $f(x)$ is differentiable at $x = -1$.

$\Rightarrow$ continuous at $x = -1$.

To check differentiability at $x = 2$;

$$\text{LHD} = f'_-(2) = \lim_{h \rightarrow 0} \frac{\frac{1}{3}(2 - h + 1)^2 - 3}{-h} = 2 \quad \& \quad \text{RHD} = f'_+(2) = \lim_{h \rightarrow 0} \frac{3(2 + h - 1)^2 - 3}{h} = 6$$

Hence $f(x)$ is not differentiable at $x = 2$.

But continuous at $x = 2$, because LHD & RHD both are finite.

$\therefore f(x)$ is continuous for all x and differentiable for all x , except $x = 2$.